

- 7 If the columns combine into $Ax = \mathbf{0}$ then each row has $r \cdot x = 0$:

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{By rows} \quad \begin{bmatrix} r_1 \cdot x \\ r_2 \cdot x \\ r_3 \cdot x \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The three rows also lie in a plane. Why is that plane perpendicular to x ?

- 8 Moving to a 4 by 4 difference equation $Ax = b$, find the four components x_1, x_2, x_3, x_4 . Then write this solution as $x = Sb$ to find the inverse matrix $S = A^{-1}$:

$$Ax = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = b.$$

- 9 What is the *cyclic* 4 by 4 difference matrix C ? It will have 1 and -1 in each row. Find all solutions $x = (x_1, x_2, x_3, x_4)$ to $Cx = \mathbf{0}$. The four columns of C lie in a “three-dimensional hyperplane” inside four-dimensional space.
- 10 A *forward* difference matrix Δ is *upper* triangular:

$$\Delta z = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} z_2 - z_1 \\ z_3 - z_2 \\ 0 - z_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = b.$$

Find z_1, z_2, z_3 from b_1, b_2, b_3 . What is the inverse matrix in $z = \Delta^{-1}b$?

- 11 Show that the forward differences $(t+1)^2 - t^2$ are $2t+1 = \text{odd numbers}$. As in calculus, the difference $(t+1)^n - t^n$ will begin with the derivative of t^n , which is _____.
- 12 The last lines of the Worked Example say that the 4 by 4 centered difference matrix in (16) is invertible. Solve $Cx = (b_1, b_2, b_3, b_4)$ to find its inverse in $x = C^{-1}b$.

Challenge Problems

- 13 The very last words say that the 5 by 5 centered difference matrix is *not* invertible. Write down the 5 equations $Cx = b$. Find a combination of left sides that gives zero. What combination of b_1, b_2, b_3, b_4, b_5 must be zero? (The 5 columns lie on a “4-dimensional hyperplane” in 5-dimensional space.)
- 14 If (a, b) is a multiple of (c, d) with $abcd \neq 0$, show that (a, c) is a multiple of (b, d) . This is surprisingly important; two columns are falling on one line. You could use numbers first to see how a, b, c, d are related. The question will lead to:

The matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has dependent columns when it has dependent rows.

Chapter 2

Solving Linear Equations

2.1 Vectors and Linear Equations

The central problem of linear algebra is to solve a system of equations. Those equations are linear, which means that the unknowns are only multiplied by numbers—we never see x times y . Our first linear system is certainly not big. But you will see how far it leads:

Two equations	$x - 2y = 1$	(1)
Two unknowns	$3x + 2y = 11$	

We begin *a row at a time*. The first equation $x - 2y = 1$ produces a straight line in the xy plane. The point $x = 1, y = 0$ is on the line because it solves that equation. The point $x = 3, y = 1$ is also on the line because $3 - 2 = 1$. If we choose $x = 101$ we find $y = 50$.

The slope of this particular line is $\frac{1}{2}$, because y increases by 1 when x changes by 2. But slopes are important in calculus and this is linear algebra!

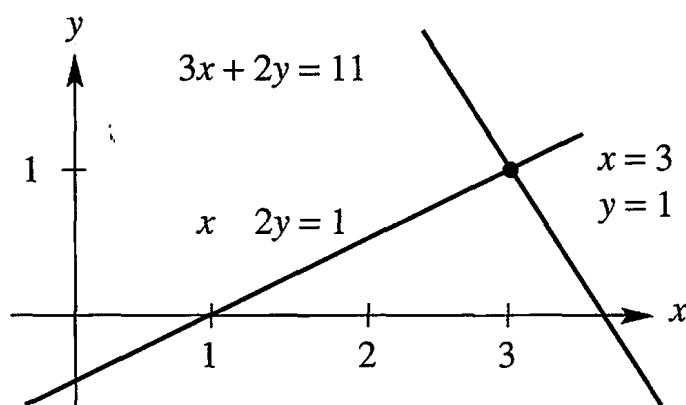


Figure 2.1: *Row picture*: The point $(3, 1)$ where the lines meet is the solution.

Figure 2.1 shows that line $x - 2y = 1$. The second line in this “row picture” comes from the second equation $3x + 2y = 11$. You can’t miss the intersection point where the

two lines meet. *The point $x = 3, y = 1$ lies on both lines.* That point solves both equations at once. This is the solution to our system of linear equations.

ROWS *The row picture shows two lines meeting at a single point (the solution).*

Turn now to the column picture. I want to recognize the same linear system as a “vector equation”. Instead of numbers we need to see *vectors*. If you separate the original system into its columns instead of its rows, you get a vector equation:

$$\text{Combination equals } b \quad x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix} = b. \quad (2)$$

This has two column vectors on the left side. The problem is *to find the combination of those vectors that equals the vector on the right*. We are multiplying the first column by x and the second column by y , and adding. With the right choices $x = 3$ and $y = 1$ (the same numbers as before), this produces $3(\text{column 1}) + 1(\text{column 2}) = b$.

COLUMNS *The column picture combines the column vectors on the left side to produce the vector b on the right side.*

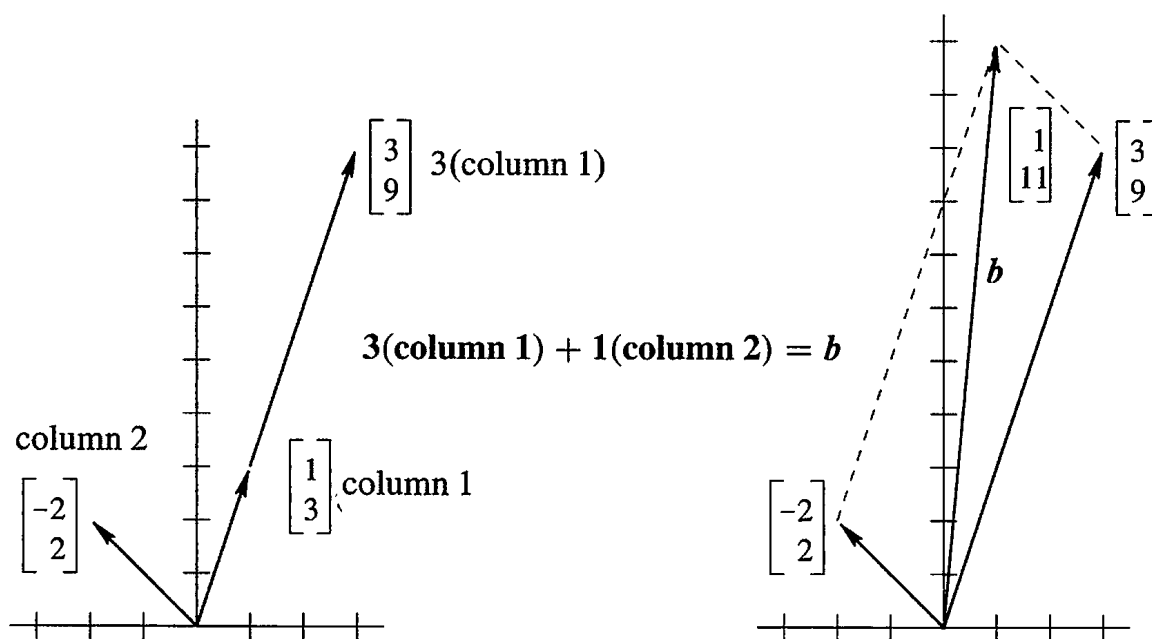


Figure 2.2: Column picture: A combination of columns produces the right side (1,11).

Figure 2.2 is the “column picture” of two equations in two unknowns. The first part shows the two separate columns, and that first column multiplied by 3. This multiplication by a *scalar* (a number) is one of the two basic operations in linear algebra:

$$\text{Scalar multiplication} \quad 3 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 9 \end{bmatrix}.$$

If the components of a vector v are v_1 and v_2 , then cv has components cv_1 and cv_2 .

The other basic operation is *vector addition*. We add the first components and the second components separately. The vector sum is $(1, 11)$ as desired:

$$\text{Vector addition} \quad \begin{bmatrix} 3 \\ 9 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}.$$

The right side of Figure 2.2 shows this addition. The sum along the diagonal is the vector $b = (1, 11)$ on the right side of the linear equations.

To repeat: The left side of the vector equation is a *linear combination* of the columns. The problem is to find the right coefficients $x = 3$ and $y = 1$. We are combining scalar multiplication and vector addition into one step. That step is crucially important, because it contains both of the basic operations:

$$\text{Linear combination} \quad 3 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}.$$

Of course the solution $x = 3, y = 1$ is the same as in the row picture. I don't know which picture you prefer! I suspect that the two intersecting lines are more familiar at first. You may like the row picture better, but only for one day. My own preference is to combine column vectors. It is a lot easier to see a combination of four vectors in four-dimensional space, than to visualize how four hyperplanes might possibly meet at a point. (*Even one hyperplane is hard enough. . .*)

The *coefficient matrix* on the left side of the equations is the 2 by 2 matrix A :

$$\text{Coefficient matrix} \quad A = \begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix}.$$

This is very typical of linear algebra, to look at a matrix by rows and by columns. Its rows give the row picture and its columns give the column picture. Same numbers, different pictures, same equations. We write those equations as a matrix problem $Ax = b$:

$$\text{Matrix equation} \quad \begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}.$$

The row picture deals with the two rows of A . The column picture combines the columns. The numbers $x = 3$ and $y = 1$ go into x . Here is matrix-vector multiplication:

$$\begin{array}{l} \text{Dot products with rows} \\ \text{Combination of columns} \end{array} \quad Ax = b \quad \text{is} \quad \begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}.$$

Looking ahead This chapter is going to solve n equations in n unknowns (for any n). I am not going at top speed, because smaller systems allow examples and pictures and a complete understanding. You are free to go faster, as long as **matrix multiplication and inversion** become clear. Those two ideas will be the keys to invertible matrices.

I can list four steps to understanding elimination using matrices.

1. Elimination goes from A to a triangular U by a sequence of matrix steps E_{ij} .
2. The inverse matrices E_{ij}^{-1} in reverse order bring U back to the original A .
3. In matrix language that reverse order is $A = LU = (\text{lower triangle}) (\text{upper triangle})$.
4. Elimination succeeds if A is invertible. (It may need row exchanges.).

The most-used algorithm in computational science takes those steps (MATLAB calls it **lu**). But linear algebra goes beyond square invertible matrices! For m by n matrices, $Ax = 0$ may have many solutions. Those solutions will go into a **vector space**. The **rank** of A leads to the **dimension** of that vector space.

All this comes in Chapter 3, and I don't want to hurry. But I must get there.

Three Equations in Three Unknowns

The three unknowns are x, y, z . We have three linear equations:

$$\begin{array}{rclcl}
 & x & + & 2y & + & 3z & = & 6 \\
 Ax = b & 2x & + & 5y & + & 2z & = & 4 \\
 & 6x & - & 3y & + & z & = & 2
 \end{array} \tag{3}$$

We look for numbers x, y, z that solve all three equations at once. Those desired numbers might or might not exist. For this system, they do exist. When the number of unknowns matches the number of equations, there is *usually* one solution. Before solving the problem, we visualize it both ways:

ROW *The row picture shows three planes meeting at a single point.*

COLUMN *The column picture combines three columns to produce $(6, 4, 2)$.*

In the row picture, each equation produces a *plane* in three-dimensional space. The first plane in Figure 2.3 comes from the first equation $x + 2y + 3z = 6$. That plane crosses the x and y and z axes at the points $(6, 0, 0)$ and $(0, 3, 0)$ and $(0, 0, 2)$. Those three points solve the equation and they determine the whole plane.

The vector $(x, y, z) = (0, 0, 0)$ does not solve $x + 2y + 3z = 6$. Therefore that plane does not contain the origin. The plane $x + 2y + 3z = 0$ does pass through the origin, and it is parallel to $x + 2y + 3z = 6$. When the right side increases to 6, the parallel plane moves away from the origin.

The second plane is given by the second equation $2x + 5y + 2z = 4$. *It intersects the first plane in a line L .* The usual result of two equations in three unknowns is a line L of solutions. (Not if the equations were $x + 2y + 3z = 6$ and $x + 2y + 3z = 0$.)

The third equation gives a third plane. It cuts the line L at a single point. That point lies on all three planes and it solves all three equations. It is harder to draw this triple intersection point than to imagine it. The three planes meet at the solution (which we haven't found yet). **The column form will now show immediately why $z = 2$.**

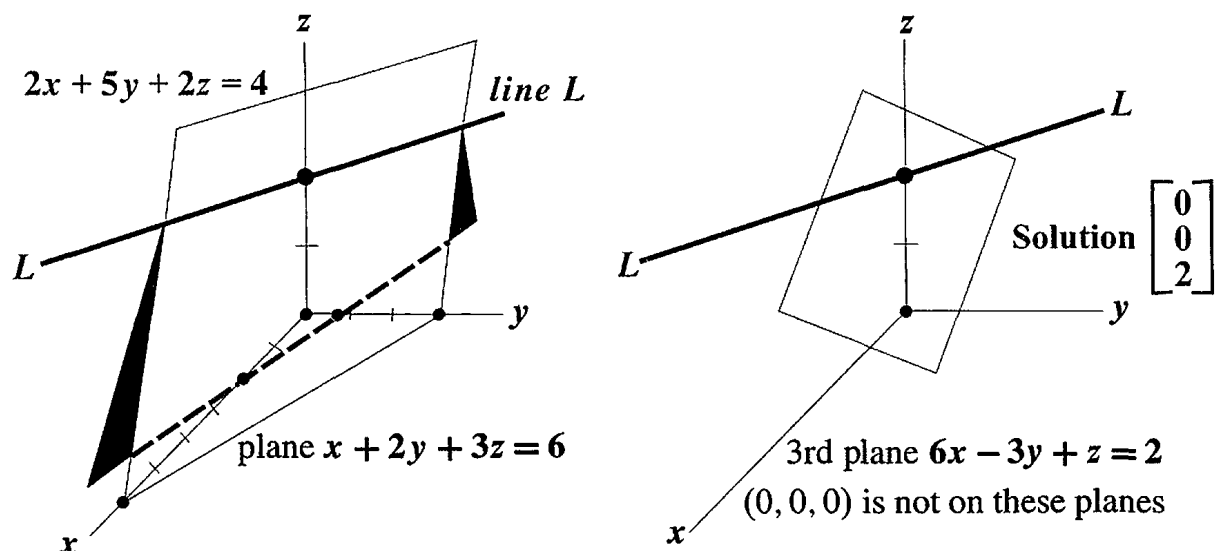


Figure 2.3: *Row picture*: Two planes meet at a line, three planes at a point.

The column picture starts with the vector form of the equations $Ax = b$:

Combine columns
$$x \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \\ -3 \end{bmatrix} + z \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}. \quad (4)$$

The unknowns are the coefficients x, y, z . We want to multiply the three column vectors by the correct numbers x, y, z to produce $\mathbf{b} = (6, 4, 2)$.

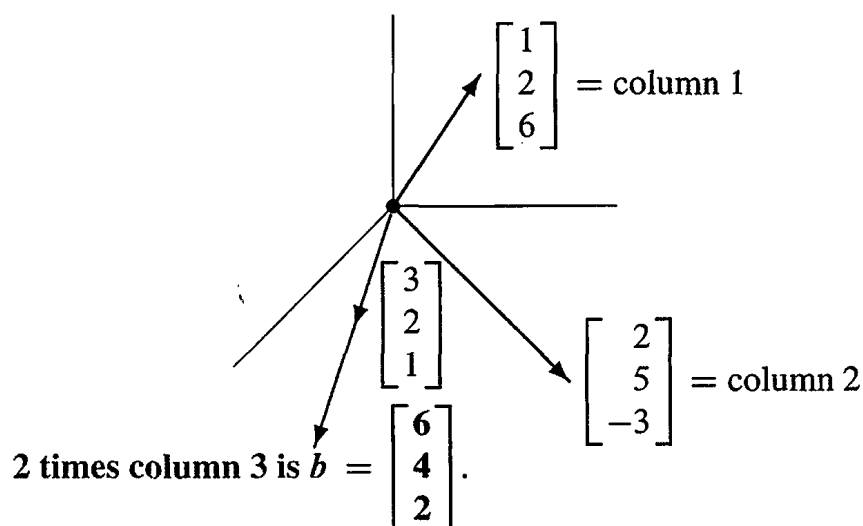


Figure 2.4: *Column picture*: $(x, y, z) = (0, 0, 2)$ because $2(3, 2, 1) = (6, 4, 2) = \mathbf{b}$.

Figure 2.4 shows this column picture. Linear combinations of those columns can produce any vector $\mathbf{b}$! The combination that produces $\mathbf{b} = (6, 4, 2)$ is just 2 times the third column. *The coefficients we need are $x = 0$, $y = 0$, and $z = 2$.*

The three planes in the row picture meet at that same solution point $(0, 0, 2)$:

Correct combination
 $(x, y, z) = (0, 0, 2)$

$$0 \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 5 \\ -3 \end{bmatrix} + 2 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}.$$

The Matrix Form of the Equations

We have three rows in the row picture and three columns in the column picture (plus the right side). The three rows and three columns contain nine numbers. *These nine numbers fill a 3 by 3 matrix A :*

The “coefficient matrix” in $Ax = b$ is

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 6 & -3 & 1 \end{bmatrix}.$$

The capital letter A stands for all nine coefficients (in this square array). The letter b denotes the column vector with components 6, 4, 2. The unknown x is also a column vector, with components x, y, z . (We use boldface because it is a vector, x because it is unknown.) By rows the equations were (3), by columns they were (4), and by matrices they are (5):

Matrix equation $Ax = b$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 6 & -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}. \quad (5)$$

Basic question: What does it mean to “multiply A times x ”? We can multiply by rows or by columns. Either way, $Ax = b$ must be a correct representation of the three equations. You do the same nine multiplications either way.

Multiplication by rows Ax comes from *dot products*, each row times the column x :

$$Ax = \begin{bmatrix} (\text{row } 1) \cdot x \\ (\text{row } 2) \cdot x \\ (\text{row } 3) \cdot x \end{bmatrix}. \quad (6)$$

Multiplication by columns Ax is a *combination of column vectors*:

$$Ax = x (\text{column } 1) + y (\text{column } 2) + z (\text{column } 3). \quad (7)$$

When we substitute the solution $x = (0, 0, 2)$, the multiplication Ax produces b :

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 6 & -3 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = 2 \text{ times column } 3 = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}.$$

The dot product from the first row is $(1, 2, 3) \cdot (0, 0, 2) = 6$. The other rows give dot products 4 and 2. *This book sees Ax as a combination of the columns of A .*

Example 1 Here are 3 by 3 matrices A and $I = \text{identity}$, with three 1's and six 0's:

$$Ax = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} \quad Ix = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

If you are a row person, the dot product of $(1, 0, 0)$ with $(4, 5, 6)$ is 4. If you are a column person, the linear combination Ax is 4 times the first column $(1, 1, 1)$. In that matrix A , the second and third columns are zero vectors.

The other matrix I is special. It has ones on the “main diagonal”. *Whatever vector this matrix multiplies, that vector is not changed.* This is like multiplication by 1, but for matrices and vectors. The exceptional matrix in this example is the 3 by 3 **identity matrix**:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{always yields the multiplication} \quad Ix = x.$$

Matrix Notation

The first row of a 2 by 2 matrix contains a_{11} and a_{12} . The second row contains a_{21} and a_{22} . The first index gives the row number, so that a_{ij} is an entry in row i . The second index j gives the column number. But those subscripts are not very convenient on a keyboard! Instead of a_{ij} we type $A(i, j)$. **The entry $a_{57} = A(5, 7)$ would be in row 5, column 7.**

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} A(1, 1) & A(1, 2) \\ A(2, 1) & A(2, 2) \end{bmatrix}.$$

For an m by n matrix, the row index i goes from 1 to m . The column index j stops at n . There are mn entries $a_{ij} = A(i, j)$. A square matrix of order n has n^2 entries.

Multiplication in MATLAB

I want to express A and x and their product Ax using MATLAB commands. This is a first step in learning that language. I begin by defining the matrix A and the vector x . This vector is a 3 by 1 matrix, with three rows and one column. Enter matrices a row at a time, and use a semicolon to signal the end of a row:

$$A = [1 \ 2 \ 3; \ 2 \ 5 \ 2; \ 6 \ -3 \ 1] \\ x = [0; 0; 2]$$

Here are three ways to multiply Ax in MATLAB. In reality, $A * x$ is the good way to do it. MATLAB is a high level language, and it works with matrices:

$$\text{Matrix multiplication} \quad b = A * x$$

We can also pick out the first row of A (as a smaller matrix!). The notation for that 1 by 3 submatrix is $A(1, :)$. **Here the colon symbol keeps all columns of row 1:**

$$\text{Row at a time} \quad b = [A(1, :) * x; A(2, :) * x; A(3, :) * x]$$

Each entry is a dot product, row times column, 1 by 3 matrix times 3 by 1 matrix.

The other way to multiply uses the columns of A . The first column is the 3 by 1 submatrix $A(:, 1)$. Now the colon symbol $:$ is keeping all rows of column 1. This column multiplies $x(1)$ and the other columns multiply $x(2)$ and $x(3)$:

$$\text{Column at a time} \quad b = A(:, 1) * x(1) + A(:, 2) * x(2) + A(:, 3) * x(3)$$

I think that matrices are stored by columns. Then multiplying a column at a time will be a little faster. So $A * x$ is actually executed by columns.

You can see the same choice in a FORTRAN-type structure, which operates on single entries of A and x . This lower level language needs an outer and inner “DO loop”. When the outer loop uses the row number I , multiplication is a row at a time. The inner loop $J = 1, 3$ goes along each row I .

When the outer loop uses J , multiplication is a column at a time. I will do that in MATLAB (which really needs two more lines “end” and “end” to close “for i ” and “for j ”).

FORTRAN by rows

```
DO 10 I = 1, 3
DO 10 J = 1, 3
10 B(I) = B(I) + A(I, J) * X(J)
```

MATLAB by columns

```
for j = 1 : 3
for i = 1 : 3
b(i) = b(i) + A(i, j) * x(j)
```

Notice that MATLAB is sensitive to upper case versus lower case (capital letters and small letters). If the matrix is A then its entries are not $a(i, j)$: not recognized.

I think you will prefer the higher level $A * x$. FORTRAN won't appear again in this book. *Maple* and *Mathematica* and graphing calculators also operate at the higher level. Multiplication is $A \cdot x$ in *Mathematica*. It is **multiply**(A, x); or equally **evalm**($A \& * x$); in *Maple*. Those languages allow symbolic entries $a, b, x, \dots$ and not only real numbers. Like MATLAB's Symbolic Toolbox, they give the symbolic answer.

■ REVIEW OF THE KEY IDEAS ■

1. The basic operations on vectors are multiplication cv and vector addition $v + w$.
2. Together those operations give linear combinations $cv + dw$.
3. Matrix-vector multiplication Ax can be computed by dot products, a row at a time. But Ax should be understood as a combination of the columns of A .
4. Column picture: $Ax = b$ asks for a combination of columns to produce b .
5. Row picture: Each equation in $Ax = b$ gives a line ($n = 2$) or a plane ($n = 3$) or a “hyperplane” ($n > 3$). They intersect at the solution or solutions, if any.

■ WORKED EXAMPLES ■

2.1 A Describe the column picture of these three equations $Ax = b$. Solve by careful inspection of the columns (instead of elimination):

$$\begin{array}{rcl} x + 3y + 2z & = & -3 \\ 2x + 2y + 2z & = & -2 \\ 3x + 5y + 6z & = & -5 \end{array} \quad \text{which is} \quad \begin{bmatrix} 1 & 3 & 2 \\ 2 & 2 & 2 \\ 3 & 5 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ -2 \\ -5 \end{bmatrix}.$$

Solution The column picture asks for a linear combination that produces b from the three columns of A . In this example b is *minus the second column*. So the solution is $x = 0, y = -1, z = 0$. To show that $(0, -1, 0)$ is the *only* solution we have to know that “ A is invertible” and “the columns are independent” and “the determinant isn’t zero.”

Those words are not yet defined but the test comes from elimination: We need (and for this matrix we find) a full set of three nonzero pivots.

Suppose the right side changes to $b = (4, 4, 8) = \text{sum of the first two columns}$. Then the good combination has $x = 1, y = 1, z = 0$. The solution becomes $x = (1, 1, 0)$.

2.1 B This system has *no solution*. The planes in the row picture don’t meet at a point. *No combination of the three columns produces b . How to show this?*

$$\begin{array}{rcl} x + 3y + 5z & = & 4 \\ x + 2y - 3z & = & 5 \\ 2x + 5y + 2z & = & 8 \end{array} \quad \begin{bmatrix} 1 & 3 & 5 \\ 1 & 2 & -3 \\ 2 & 5 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 8 \end{bmatrix} = b$$

- (1) Multiply the equations by 1, 1, -1 and add to get $0 = 1$. *No solution*. Are any two of the planes parallel? What are the equations of planes parallel to $x + 3y + 5z = 4$?
- (2) Take the dot product of each column of A (and also b) with $y = (1, 1, -1)$. How do those dot products show that the system $Ax = b$ has no solution?
- (3) Find three right side vectors b^* and b^{**} and b^{***} that *do* allow solutions.

Solution

- (1) Multiplying the equations by 1, 1, -1 and adding gives $0 = 1$:

$$\begin{array}{rcl} x + 3y + 5z & = & 4 \\ x + 2y - 3z & = & 5 \\ -[2x + 5y + 2z = 8] & & \\ \hline 0x + 0y + 0z & = & 1 \quad \text{No Solution} \end{array}$$

The planes don’t meet at a point, even though no two planes are parallel. For a plane parallel to $x + 3y + 5z = 4$, change the “4”. The parallel plane $x + 3y + 5z = 0$ goes through the origin $(0, 0, 0)$. And the equation multiplied by any nonzero constant still gives the same plane, as in $2x + 6y + 10z = 8$.

- (2) The dot product of each column of A with $y = (1, 1, -1)$ is *zero*. On the right side, $y \cdot b = (1, 1, -1) \cdot (4, 5, 8) = 1$ is *not zero*. So a solution is impossible.
- (3) There is a solution when b is a combination of the columns. These three choices of b have solutions $x^* = (1, 0, 0)$ and $x^{**} = (1, 1, 1)$ and $x^{***} = (0, 0, 0)$:

$$b^* = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \text{first column} \quad b^{**} = \begin{bmatrix} 9 \\ 0 \\ 9 \end{bmatrix} = \text{sum of columns} \quad b^{***} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Problem Set 2.1

Problems 1–8 are about the row and column pictures of $Ax = b$.

- 1 With $A = I$ (the identity matrix) draw the planes in the row picture. Three sides of a box meet at the solution $x = (x, y, z) = (2, 3, 4)$:

$$\begin{array}{l} 1x + 0y + 0z = 2 \\ 0x + 1y + 0z = 3 \\ 0x + 0y + 1z = 4 \end{array} \quad \text{or} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

Draw the vectors in the column picture. Two times column 1 plus three times column 2 plus four times column 3 equals the right side b .

- 2 If the equations in Problem 1 are multiplied by 2, 3, 4 they become $DX = B$:

$$\begin{array}{l} 2x + 0y + 0z = 4 \\ 0x + 3y + 0z = 9 \\ 0x + 0y + 4z = 16 \end{array} \quad \text{or} \quad DX = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 16 \end{bmatrix} = B$$

Why is the row picture the same? Is the solution X the same as x ? What is changed in the column picture—the columns or the right combination to give B ?

- 3 If equation 1 is added to equation 2, which of these are changed: the planes in the row picture, the vectors in the column picture, the coefficient matrix, the solution? The new equations in Problem 1 would be $x = 2$, $x + y = 5$, $z = 4$.
- 4 Find a point with $z = 2$ on the intersection line of the planes $x + y + 3z = 6$ and $x - y + z = 4$. Find the point with $z = 0$. Find a third point halfway between.
- 5 The first of these equations plus the second equals the third:

$$\begin{array}{l} x + y + z = 2 \\ x + 2y + z = 3 \\ 2x + 3y + 2z = 5. \end{array}$$

The first two planes meet along a line. The third plane contains that line, because if x, y, z satisfy the first two equations then they also _____. The equations have infinitely many solutions (the whole line L). Find three solutions on L .

- 6 Move the third plane in Problem 5 to a parallel plane $2x + 3y + 2z = 9$. Now the three equations have no solution—*why not*? The first two planes meet along the line L , but the third plane doesn't _____ that line.
- 7 In Problem 5 the columns are $(1, 1, 2)$ and $(1, 2, 3)$ and $(1, 1, 2)$. This is a “singular case” because the third column is _____. Find two combinations of the columns that give $\mathbf{b} = (2, 3, 5)$. This is only possible for $\mathbf{b} = (4, 6, c)$ if $c = \underline{\hspace{1cm}}$.
- 8 Normally 4 “planes” in 4-dimensional space meet at a _____. Normally 4 column vectors in 4-dimensional space can combine to produce $\mathbf{b}$. What combination of $(1, 0, 0, 0)$, $(1, 1, 0, 0)$, $(1, 1, 1, 0)$, $(1, 1, 1, 1)$ produces $\mathbf{b} = (3, 3, 3, 2)$? What 4 equations for x, y, z, t are you solving?

Problems 9–14 are about multiplying matrices and vectors.

- 9 Compute each $A\mathbf{x}$ by dot products of the rows with the column vector:

$$(a) \begin{bmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \quad (b) \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

- 10 Compute each $A\mathbf{x}$ in Problem 9 as a combination of the columns:

$$9(a) \text{ becomes } A\mathbf{x} = 2 \begin{bmatrix} 1 \\ -2 \\ -4 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix}.$$

How many separate multiplications for $A\mathbf{x}$, when the matrix is “3 by 3”?

- 11 Find the two components of $A\mathbf{x}$ by rows or by columns:

$$\begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}.$$

- 12 Multiply A times $\mathbf{x}$ to find three components of $A\mathbf{x}$:

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \\ 3 & 3 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

- 13 (a) A matrix with m rows and n columns multiplies a vector with _____ components to produce a vector with _____ components.
- (b) The planes from the m equations $A\mathbf{x} = \mathbf{b}$ are in _____-dimensional space. The combination of the columns of A is in _____-dimensional space.

- 14 Write $2x + 3y + z + 5t = 8$ as a matrix A (how many rows?) multiplying the column vector $\mathbf{x} = (x, y, z, t)$ to produce $\mathbf{b}$. The solutions $\mathbf{x}$ fill a plane or “hyperplane” in 4-dimensional space. *The plane is 3-dimensional with no 4D volume.*

Problems 15–22 ask for matrices that act in special ways on vectors.

- 15 (a) What is the 2 by 2 identity matrix? I times $\begin{bmatrix} x \\ y \end{bmatrix}$ equals $\begin{bmatrix} x \\ y \end{bmatrix}$.
 (b) What is the 2 by 2 exchange matrix? P times $\begin{bmatrix} x \\ y \end{bmatrix}$ equals $\begin{bmatrix} y \\ x \end{bmatrix}$.
- 16 (a) What 2 by 2 matrix R rotates every vector by 90° ? R times $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} -y \\ x \end{bmatrix}$.
 (b) What 2 by 2 matrix R^2 rotates every vector by 180° ?
- 17 Find the matrix P that multiplies (x, y, z) to give (y, z, x) . Find the matrix Q that multiplies (y, z, x) to bring back (x, y, z) .
- 18 What 2 by 2 matrix E subtracts the first component from the second component? What 3 by 3 matrix does the same?

$$E \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \text{and} \quad E \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 7 \end{bmatrix}.$$

- 19 What 3 by 3 matrix E multiplies (x, y, z) to give $(x, y, z + x)$? What matrix E^{-1} multiplies (x, y, z) to give $(x, y, z - x)$? If you multiply $(3, 4, 5)$ by E and then multiply by E^{-1} , the two results are (____) and (____).
- 20 What 2 by 2 matrix P_1 projects the vector (x, y) onto the x axis to produce $(x, 0)$? What matrix P_2 projects onto the y axis to produce $(0, y)$? If you multiply $(5, 7)$ by P_1 and then multiply by P_2 , you get (____) and (____).
- 21 What 2 by 2 matrix R rotates every vector through 45° ? The vector $(1, 0)$ goes to $(\sqrt{2}/2, \sqrt{2}/2)$. The vector $(0, 1)$ goes to $(-\sqrt{2}/2, \sqrt{2}/2)$. Those determine the matrix. Draw these particular vectors in the xy plane and find R .
- 22 Write the dot product of $(1, 4, 5)$ and (x, y, z) as a matrix multiplication $A\mathbf{x}$. The matrix A has one row. The solutions to $A\mathbf{x} = \mathbf{0}$ lie on a ____ perpendicular to the vector _____. The columns of A are only in ____-dimensional space.
- 23 In MATLAB notation, write the commands that define this matrix A and the column vectors $\mathbf{x}$ and $\mathbf{b}$. What command would test whether or not $A\mathbf{x} = \mathbf{b}$?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} 5 \\ -2 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

- 24 The MATLAB commands $A = \text{eye}(3)$ and $\mathbf{v} = [3:5]'$ produce the 3 by 3 identity matrix and the column vector $(3, 4, 5)$. What are the outputs from $A*\mathbf{v}$ and $\mathbf{v}'*\mathbf{v}$? (Computer not needed!) If you ask for $\mathbf{v}*A$, what happens?

- 25** If you multiply the 4 by 4 all-ones matrix $A = \text{ones}(4)$ and the column $v = \text{ones}(4,1)$, what is $A*v$? (Computer not needed.) If you multiply $B = \text{eye}(4) + \text{ones}(4)$ times $w = \text{zeros}(4,1) + 2*\text{ones}(4,1)$, what is $B*w$?

Questions 26–28 review the row and column pictures in 2, 3, and 4 dimensions.

- 26** Draw the row and column pictures for the equations $x - 2y = 0$, $x + y = 6$.
- 27** For two linear equations in three unknowns x, y, z , the row picture will show (2 or 3) (lines or planes) in (2 or 3)-dimensional space. The column picture is in (2 or 3)-dimensional space. The solutions normally lie on a _____.
- 28** For four linear equations in two unknowns x and y , the row picture shows four _____. The column picture is in _____-dimensional space. The equations have no solution unless the vector on the right side is a combination of _____.
- 29** Start with the vector $u_0 = (1, 0)$. Multiply again and again by the same “Markov matrix” $A = [.8 \ .3; .2 \ .7]$. The next three vectors are u_1, u_2, u_3 :

$$u_1 = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} .8 \\ .2 \end{bmatrix} \quad u_2 = Au_1 = \quad \quad u_3 = Au_2 = \quad .$$

What property do you notice for all four vectors u_0, u_1, u_2, u_3 ?

Challenge Problems

- 30** Continue Problem 29 from $u_0 = (1, 0)$ to u_7 , and also from $v_0 = (0, 1)$ to v_7 . What do you notice about u_7 and v_7 ? Here are two MATLAB codes, with while and for. They plot u_0 to u_7 and v_0 to v_7 . You can use other languages:

```
u = [1 ; 0]; A = [.8 .3 ; .2 .7];
x = u; k = [0 : 7];
while size(x,2) <= 7
    u = A*u; x = [x u];
end
plot(k, x)
```

```
v = [0 ; 1]; A = [.8 .3 ; .2 .7];
x = v; k = [0 : 7];
for j = 1 : 7
    v = A*v; x = [x v];
end
plot(k, x)
```

The u 's and v 's are approaching a steady state vector s . Guess that vector and check that $As = s$. If you start with s , you stay with s .

- 31** Invent a 3 by 3 **magic matrix** M_3 with entries $1, 2, \dots, 9$. All rows and columns and diagonals add to 15. The first row could be 8, 3, 4. What is M_3 times $(1, 1, 1)$? What is M_4 times $(1, 1, 1, 1)$ if a 4 by 4 magic matrix has entries $1, \dots, 16$?
- 32** Suppose u and v are the first two columns of a 3 by 3 matrix A . Which third columns w would make this matrix singular? Describe a typical column picture of $Ax = b$ in that singular case, and a typical row picture (for a random b).

33 Multiplying by A is a “linear transformation”. Those important words mean:

If w is a combination of u and v , then Aw is the same combination of Au and Av .

It is this “*linearity*” $Aw = cAu + dAv$ that gives us the name *linear algebra*.

Problem: If $u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ then Au and Av are the columns of A .

Combine $w = cu + dv$. If $w = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$ how is Aw connected to Au and Av ?

34 Start from the four equations $-x_{i+1} + 2x_i - x_{i-1} = i$ (for $i = 1, 2, 3, 4$ with $x_0 = x_5 = 0$). Write those equations in their matrix form $Ax = b$. Can you solve them for x_1, x_2, x_3, x_4 ?

35 A 9 by 9 *Sudoku matrix* S has the numbers $1, \dots, 9$ in every row and column, and in every 3 by 3 block. For the all-ones vector $x = (1, \dots, 1)$, what is Sx ?

A better question is: **Which row exchanges will produce another Sudoku matrix?** Also, which exchanges of block rows give another Sudoku matrix?

Section 2.7 will look at all possible permutations (reorderings) of the rows. I can see 6 orders for the first 3 rows, all giving Sudoku matrices. Also 6 permutations of the next 3 rows, and of the last 3 rows. And 6 block permutations of the block rows?

2.2 The Idea of Elimination

This chapter explains a systematic way to solve linear equations. The method is called “*elimination*”, and you can see it immediately in our 2 by 2 example. Before elimination, x and y appear in both equations. After elimination, the first unknown x has disappeared from the second equation $8y = 8$:

Before	$\begin{array}{rcl} x - 2y & = & 1 \\ 3x + 2y & = & 11 \end{array}$	After	$\begin{array}{rcl} x - 2y & = & 1 \\ 8y & = & 8 \end{array}$	(multiply equation 1 by 3) (subtract to eliminate $3x$)
---------------	---	--------------	---	---

The new equation $8y = 8$ instantly gives $y = 1$. Substituting $y = 1$ back into the first equation leaves $x - 2 = 1$. Therefore $x = 3$ and the solution $(x, y) = (3, 1)$ is complete.

Elimination produces an *upper triangular system*—this is the goal. The nonzero coefficients $1, -2, 8$ form a triangle. That system is solved from the bottom upwards—first $y = 1$ and then $x = 3$. This quick process is called *back substitution*. It is used for upper triangular systems of any size, after elimination gives a triangle.

Important point: The original equations have the same solution $x = 3$ and $y = 1$. Figure 2.5 shows each system as a pair of lines, intersecting at the solution point $(3, 1)$. After elimination, the lines still meet at the same point. Every step worked with correct equations.

How did we get from the first pair of lines to the second pair? We subtracted 3 times the first equation from the second equation. The step that eliminates x from equation 2 is the fundamental operation in this chapter. We use it so often that we look at it closely:

To eliminate x : Subtract a multiple of equation 1 from equation 2.

Three times $x - 2y = 1$ gives $3x - 6y = 3$. When this is subtracted from $3x + 2y = 11$, the right side becomes 8. The main point is that $3x$ cancels $3x$. What remains on the left side is $2y - (-6y)$ or $8y$, and x is eliminated. **The system became triangular.**

Ask yourself how that multiplier $\ell = 3$ was found. The first equation contains $1x$. **So the first pivot was 1** (the coefficient of x). The second equation contains $3x$, **so the multiplier was 3**. Then subtraction $3x - 3x$ produced the zero and the triangle.

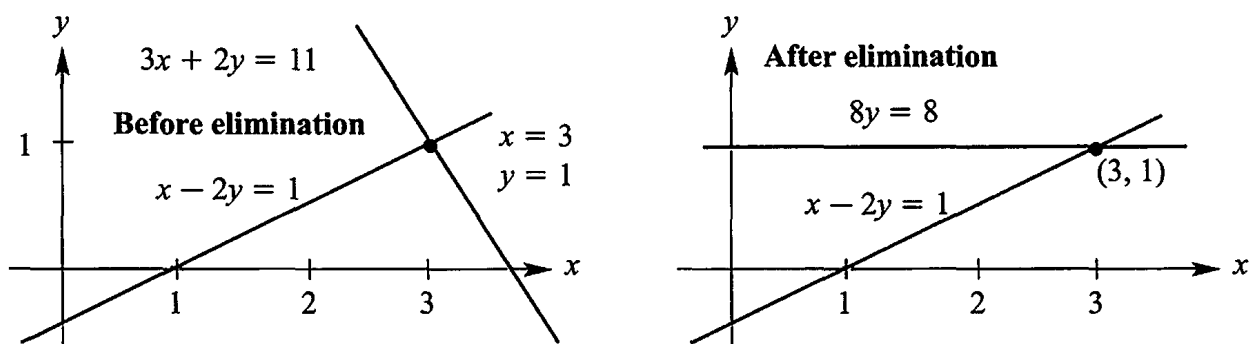


Figure 2.5: Eliminating x makes the second line horizontal. Then $8y = 8$ gives $y = 1$.

You will see the multiplier rule if I change the first equation to $4x - 8y = 4$. (Same straight line but the first pivot becomes 4.) The correct multiplier is now $\ell = \frac{3}{4}$. *To find the multiplier, divide the coefficient “3” to be eliminated by the pivot “4”:*

$$\begin{array}{rcl} 4x - 8y = 4 & \text{Multiply equation 1 by } \frac{3}{4} & \triangleleft 4x - 8y = 4 \\ 3x + 2y = 11 & \text{Subtract from equation 2} & \quad 8y = 8. \end{array}$$

The final system is triangular and the last equation still gives $y = 1$. Back substitution produces $4x - 8 = 4$ and $4x = 12$ and $x = 3$. We changed the numbers but not the lines or the solution. *Divide by the pivot to find that multiplier $\ell = \frac{3}{4}$:*

$$\begin{aligned} \text{Pivot} &= \text{first nonzero in the row that does the elimination} \\ \text{Multiplier} &= (\text{entry to eliminate}) \text{ divided by (pivot)} = \frac{3}{4}. \end{aligned}$$

The new second equation starts with the second pivot, which is 8. We would use it to eliminate y from the third equation if there were one. *To solve n equations we want n pivots. The pivots are on the diagonal of the triangle after elimination.*

You could have solved those equations for x and y without reading this book. It is an extremely humble problem, but we stay with it a little longer. Even for a 2 by 2 system, elimination might break down. By understanding the possible breakdown (when we can't find a full set of pivots), you will understand the whole process of elimination.

Breakdown of Elimination

Normally, elimination produces the pivots that take us to the solution. But failure is possible. At some point, the method might ask us to *divide by zero*. We can't do it. The process has to stop. There might be a way to adjust and continue—or failure may be unavoidable.

Example 1 fails with *no solution* to $0y = 8$. Example 2 fails with *too many solutions* to $0y = 0$. Example 3 succeeds by exchanging the equations.

Example 1 *Permanent failure with no solution.* Elimination makes this clear:

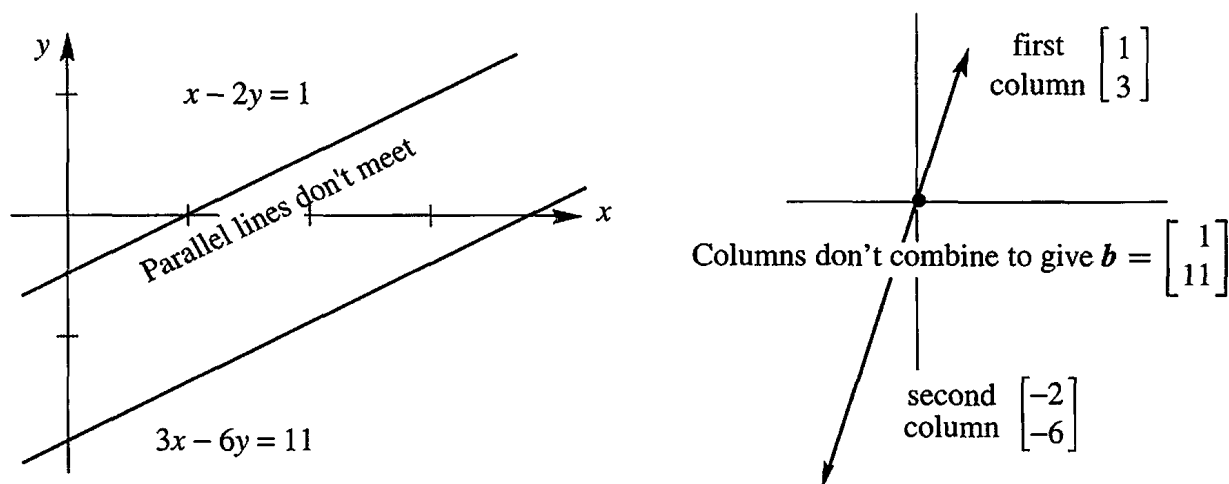
$$\begin{array}{rcl} x - 2y = 1 & \text{Subtract 3 times} & x - 2y = 1 \\ 3x - 6y = 11 & \text{eqn. 1 from eqn. 2} & 0y = 8. \end{array}$$

There is *no* solution to $0y = 8$. Normally we divide the right side 8 by the second pivot, but *this system has no second pivot. (Zero is never allowed as a pivot!)* The row and column pictures in Figure 2.6 show why failure was unavoidable. If there is no solution, elimination will discover that fact by reaching an equation like $0y = 8$.

The row picture of failure shows parallel lines—which never meet. A solution must lie on both lines. With no meeting point, the equations have no solution.

The column picture shows the two columns $(1, 3)$ and $(-2, -6)$ in the same direction. *All combinations of the columns lie along a line.* But the column from the right side is in a different direction $(1, 11)$. No combination of the columns can produce this right side—therefore no solution.

When we change the right side to $(1, 3)$, failure shows as a whole line of solution points. Instead of no solution, next comes Example 2 with infinitely many.

Figure 2.6: Row picture and column picture for Example 1: *no solution*.

Example 2 *Failure with infinitely many solutions. Change $b = (1, 11)$ to $(1, 3)$.*

$x - 2y = 1$	Subtract 3 times	$x - 2y = 1$	Still only
$3x - 6y = 3$	eqn. 1 from eqn. 2	$0y = 0$	one pivot.

Every y satisfies $0y = 0$. There is really only one equation $x - 2y = 1$. The unknown y is “free”. After y is freely chosen, x is determined as $x = 1 + 2y$.

In the row picture, the parallel lines have become the same line. Every point on that line satisfies both equations. We have a whole line of solutions in Figure 2.7.

In the column picture, $b = (1, 3)$ is now the same as column 1. So we can choose $x = 1$ and $y = 0$. We can also choose $x = 0$ and $y = -\frac{1}{2}$; column 2 times $-\frac{1}{2}$ equals b . Every (x, y) that solves the row problem also solves the column problem.

Failure For n equations we do not get n pivots

Elimination leads to an equation $0 \neq 0$ (no solution) or **$0 = 0$** (many solutions)

Success comes with n pivots. But we may have to exchange the n equations.

Elimination can go wrong in a third way—but this time it can be fixed. Suppose the first pivot position contains zero. We refuse to allow zero as a pivot. When the first equation has no term involving x , we can exchange it with an equation below:

Example 3 *Temporary failure (zero in pivot). A row exchange produces two pivots:*

Permutation	$0x + 2y = 4$	Exchange the two equations	$3x - 2y = 5$
	$3x - 2y = 5$		$2y = 4$

The new system is already triangular. This small example is ready for back substitution. The last equation gives $y = 2$, and then the first equation gives $x = 3$. The row picture is normal (two intersecting lines). The column picture is also normal (column vectors not in the same direction). The pivots 3 and 2 are normal—but a **row exchange** was required.

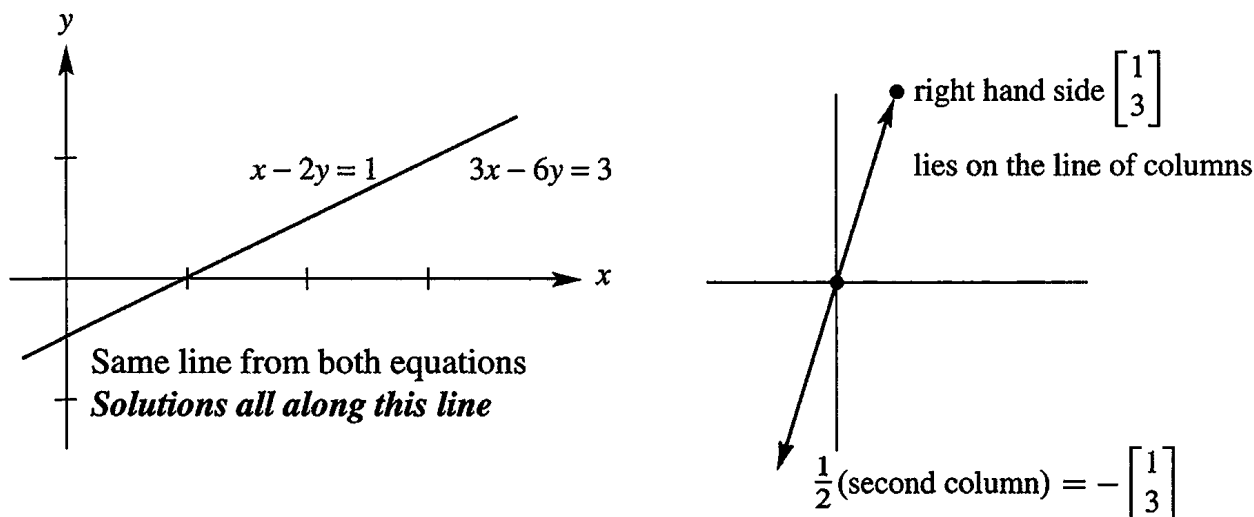


Figure 2.7: Row and column pictures for Example 2: *infinitely many solutions*.

Examples 1 and 2 are *singular*—there is no second pivot. Example 3 is *nonsingular*—there is a full set of pivots and exactly one solution. Singular equations have no solution or infinitely many solutions. Pivots must be nonzero because we have to divide by them.

Three Equations in Three Unknowns

To understand Gaussian elimination, you have to go beyond 2 by 2 systems. Three by three is enough to see the pattern. For now the matrices are square—an equal number of rows and columns. Here is a 3 by 3 system, specially constructed so that all steps lead to whole numbers and not fractions:

$$\begin{aligned} 2x + 4y - 2z &= 2 \\ 4x + 9y - 3z &= 8 \\ -2x - 3y + 7z &= 10 \end{aligned} \tag{1}$$

What are the steps? The first pivot is the boldface 2 (upper left). Below that pivot we want to eliminate the 4. *The first multiplier is the ratio $4/2 = 2$* . Multiply the pivot equation by $\ell_{21} = 2$ and subtract. Subtraction removes the $4x$ from the second equation:

Step 1 Subtract 2 times equation 1 from equation 2. This leaves $y + z = 4$.

We also eliminate $-2x$ from equation 3—still using the first pivot. The quick way is to add equation 1 to equation 3. Then $2x$ cancels $-2x$. We do exactly that, but the rule in this book is to *subtract rather than add*. The systematic pattern has multiplier $\ell_{31} = -2/2 = -1$. Subtracting -1 times an equation is the same as adding:

Step 2 Subtract -1 times equation 1 from equation 3. This leaves $y + 5z = 12$.

The two new equations involve only y and z . The second pivot (in boldface) is 1:

$$\begin{aligned} x \text{ is eliminated} \quad & \mathbf{1}y + 1z = 4 \\ & 1y + 5z = 12 \end{aligned}$$

We have reached a 2 by 2 system. The final step eliminates y to make it 1 by 1:

Step 3 Subtract equation 2_{new} from 3_{new}. The multiplier is $1/1 = 1$. Then $4z = 8$. The original $Ax = b$ has been converted into an upper triangular $Ux = c$:

$$\begin{array}{ccc}
 2x + 4y - 2z = 2 & Ax = b & 2x + 4y - 2z = 2 \\
 4x + 9y - 3z = 8 & \text{has become} & \begin{array}{c} 1y + 1z = 4 \\ 4z = 8. \end{array} \\
 -2x - 3y + 7z = 10 & Ux = c &
 \end{array} \quad (2)$$

The goal is achieved—forward elimination is complete from A to U . *Notice the pivots 2, 1, 4 along the diagonal of U .* The pivots 1 and 4 were hidden in the original system. Elimination brought them out. $Ux = c$ is ready for *back substitution*, which is quick:

$$(4z = 8 \text{ gives } z = 2) \quad (y + z = 4 \text{ gives } y = 2) \quad (\text{equation 1 gives } x = -1)$$

The solution is $(x, y, z) = (-1, 2, 2)$. The row picture has three planes from three equations. All the planes go through this solution. The original planes are sloping, but the last plane $4z = 8$ after elimination is horizontal.

The column picture shows a combination Ax of column vectors producing the right side b . The coefficients in that combination are $-1, 2, 2$ (the solution):

$$Ax = (-1) \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 9 \\ -3 \end{bmatrix} + 2 \begin{bmatrix} -2 \\ -3 \\ 7 \end{bmatrix} \text{ equals } \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix} = b. \quad (3)$$

The numbers x, y, z multiply columns 1, 2, 3 in $Ax = b$ and also in the triangular $Ux = c$.

For a 4 by 4 problem, or an n by n problem, elimination proceeds the same way. Here is the whole idea, column by column from A to U , when elimination succeeds.

Column 1. Use the first equation to create zeros below the first pivot.

Column 2. Use the new equation 2 to create zeros below the second pivot.

Columns 3 to n . Keep going to find all n pivots and the triangular U .

$$\text{After column 2 we have } \begin{bmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & x & x \end{bmatrix}. \quad \text{We want } \begin{bmatrix} x & x & x & x \\ & x & x & x \\ & & x & x \\ & & & x \end{bmatrix}. \quad (4)$$

The result of forward elimination is an upper triangular system. It is nonsingular if there is a full set of n pivots (never zero!). *Question:* Which x on the left could be changed to boldface x because the pivot is known? Here is a final example to show the original $Ax = b$, the triangular system $Ux = c$, and the solution (x, y, z) from back substitution:

$$\begin{array}{llll}
 x + y + z = 6 & & x + y + z = 6 & \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \quad \text{Back} \\
 x + 2y + 2z = 9 & \text{Forward} & y + z = 3 & \text{Back} \\
 x + 2y + 3z = 10 & \text{Forward} & z = 1 &
 \end{array}$$

All multipliers are 1. All pivots are 1. All planes meet at the solution $(3, 2, 1)$. The columns of A combine with 3, 2, 1 to give $b = (6, 9, 10)$. The triangle shows $Ux = c = (6, 3, 1)$.

■ REVIEW OF THE KEY IDEAS ■

1. A linear system ($Ax = b$) becomes upper triangular ($Ux = c$) after elimination.
2. We subtract ℓ_{ij} times equation j from equation i , to make the (i, j) entry zero.
3. The multiplier is $\ell_{ij} = \frac{\text{entry to eliminate in row } i}{\text{pivot in row } j}$. Pivots can not be zero!
4. A zero in the pivot position can be repaired if there is a nonzero below it.
5. The upper triangular system is solved by back substitution (starting at the bottom).
6. When breakdown is permanent, the system has no solution or infinitely many.

■ WORKED EXAMPLES ■

2.2 A When elimination is applied to this matrix A , what are the first and second pivots? What is the multiplier ℓ_{21} in the first step (ℓ_{21} times row 1 is *subtracted* from row 2)?

A has a *first difference* in row 1 and a *second difference* $-1, 2, -1$ in row 2.

$$A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 2 \end{bmatrix} \longrightarrow U = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

What entry in the 2, 2 position (instead of 2) would force an exchange of rows 2 and 3? Why is the lower left multiplier $\ell_{31} = 0$, subtracting zero times row 1 from row 3?

If you change the corner entry from $a_{33} = 2$ to $a_{33} = 1$, why does elimination fail?

Solution The first pivot is 1. The multiplier ℓ_{21} is $-1/1 = -1$. When -1 times row 1 is subtracted (so row 1 is added to row 2), the second pivot is revealed as 1.

If we reduce the middle entry “2” to “1”, that would force a row exchange. (Zero will appear in the second pivot position.) The multiplier ℓ_{31} is zero because $a_{31} = 0$. A zero at the start of a row needs no elimination. This A is a “*band matrix*”.

The last pivot is 1. So if the original corner entry a_{33} is reduced by 1 (to $a_{33} = 1$), elimination would produce 0. **No third pivot, elimination fails.**

2.2 B Suppose A is already a *triangular matrix* (upper triangular or lower triangular). Where do you see its pivots? When does $Ax = b$ have exactly one solution for every b ?

Solution The pivots of a triangular matrix are already set along the main diagonal. *Elimination succeeds when all those numbers are nonzero.* Use *back* substitution when A is upper triangular, go *forward* when A is lower triangular.

2.2 C Use elimination to reach upper triangular matrices U . Solve by back substitution or explain why this is impossible. What are the pivots (never zero)? Exchange equations when necessary. The only difference is the $-x$ in the last equation.

Success	$x + y + z = 7$	$x + y + z = 7$
then	$x + y - z = 5$	$x + y - z = 5$
Failure	$x - y + z = 3$	$-x - y + z = 3$

Solution For the first system, subtract equation 1 from equations 2 and 3 (the multipliers are $\ell_{21} = 1$ and $\ell_{31} = 1$). The 2, 2 entry becomes zero, so exchange equations:

	$x + y + z = 7$		$x + y + z = 7$
Success	$0y - 2z = -2$	exchanges into	$-2y + 0z = -4$
	$-2y + 0z = -4$		$-2z = -2$

Then back substitution gives $z = 1$ and $y = 2$ and $x = 4$. The pivots are 1, -2 , -2 .

For the second system, subtract equation 1 from equation 2 as before. Add equation 1 to equation 3. This leaves zero in the 2, 2 entry *and also below*:

	$x + y + z = 7$		There is <i>no pivot</i> in column 2 (it was $-$ column 1)
Failure	$0y - 2z = -2$		A further elimination step gives $0z = 8$
	$0y + 2z = 10$		The three planes don't meet

Plane 1 meets plane 2 in a line. Plane 1 meets plane 3 in a parallel line. *No solution.*

If we change the “3” in the original third equation to “ -5 ” then elimination would lead to $0 = 0$. There are infinitely many solutions! *The three planes now meet along a whole line.*

Changing 3 to -5 moved the third plane to meet the other two. The second equation gives $z = 1$. Then the first equation leaves $x + y = 6$. **No pivot in column 2 makes y free** (it can have any value). Then $x = 6 - y$.

Problem Set 2.2

Problems 1–10 are about elimination on 2 by 2 systems.

- 1 What multiple ℓ_{21} of equation 1 should be subtracted from equation 2?

$$\begin{aligned} 2x + 3y &= 1 \\ 10x + 9y &= 11. \end{aligned}$$

After this elimination step, write down the upper triangular system and circle the two pivots. The numbers 1 and 11 have no influence on those pivots.

- 2 Solve the triangular system of Problem 1 by back substitution, y before x . Verify that x times (2, 10) plus y times (3, 9) equals (1, 11). If the right side changes to (4, 44), what is the new solution?

- 3 What multiple of equation 1 should be *subtracted* from equation 2?

$$2x - 4y = 6$$

$$-x + 5y = 0.$$

After this elimination step, solve the triangular system. If the right side changes to $(-6, 0)$, what is the new solution?

- 4 What multiple ℓ of equation 1 should be subtracted from equation 2 to remove c ?

$$ax + by = f$$

$$cx + dy = g.$$

The first pivot is a (assumed nonzero). Elimination produces what formula for the second pivot? What is y ? The second pivot is missing when $ad = bc$: singular.

- 5 Choose a right side which gives no solution and another right side which gives infinitely many solutions. What are two of those solutions?

$$\begin{array}{l} \text{Singular system} \quad 3x + 2y = 10 \\ \quad \quad \quad \quad \quad 6x + 4y = \end{array}$$

- 6 Choose a coefficient b that makes this system singular. Then choose a right side g that makes it solvable. Find two solutions in that singular case.

$$2x + by = 16$$

$$4x + 8y = g.$$

- 7 For which numbers a does elimination break down (1) permanently (2) temporarily?

$$ax + 3y = -3$$

$$4x + 6y = 6.$$

Solve for x and y after fixing the temporary breakdown by a row exchange.

- 8 For which three numbers k does elimination break down? Which is fixed by a row exchange? In each case, is the number of solutions 0 or 1 or ∞ ?

$$kx + 3y = 6$$

$$3x + ky = -6.$$

- 9 What test on b_1 and b_2 decides whether these two equations allow a solution? How many solutions will they have? Draw the column picture for $\mathbf{b} = (1, 2)$ and $(1, 0)$.

$$3x - 2y = b_1$$

$$6x - 4y = b_2.$$

- 10 In the xy plane, draw the lines $x + y = 5$ and $x + 2y = 6$ and the equation $y = \underline{\hspace{1cm}}$ that comes from elimination. The line $5x - 4y = c$ will go through the solution of these equations if $c = \underline{\hspace{1cm}}$.

Problems 11–20 study elimination on 3 by 3 systems (and possible failure).

- 11** (Recommended) A system of linear equations can't have exactly two solutions. *Why?*

- (a) If (x, y, z) and (X, Y, Z) are two solutions, what is another solution?
 (b) If 25 planes meet at two points, where else do they meet?

- 12** Reduce this system to upper triangular form by two row operations:

$$\begin{aligned} 2x + 3y + z &= 8 \\ 4x + 7y + 5z &= 20 \\ -2y + 2z &= 0. \end{aligned}$$

Circle the pivots. Solve by back substitution for z, y, x .

- 13** Apply elimination (circle the pivots) and back substitution to solve

$$\begin{aligned} 2x - 3y &= 3 \\ 4x - 5y + z &= 7 \\ 2x - y - 3z &= 5. \end{aligned}$$

List the three row operations: Subtract _____ times row _____ from row _____.

- 14** Which number d forces a row exchange, and what is the triangular system (not singular) for that d ? Which d makes this system singular (no third pivot)?

$$\begin{aligned} 2x + 5y + z &= 0 \\ 4x + dy + z &= 2 \\ y - z &= 3. \end{aligned}$$

- 15** Which number b leads later to a row exchange? Which b leads to a missing pivot? In that singular case find a nonzero solution x, y, z .

$$\begin{aligned} x + by &= 0 \\ x - 2y - z &= 0 \\ y + z &= 0. \end{aligned}$$

- 16** (a) Construct a 3 by 3 system that needs two row exchanges to reach a triangular form and a solution.
 (b) Construct a 3 by 3 system that needs a row exchange to keep going, but breaks down later.
- 17** If rows 1 and 2 are the same, how far can you get with elimination (allowing row exchange)? If columns 1 and 2 are the same, which pivot is missing?

Equal	$2x - y + z = 0$	$2x + 2y + z = 0$	Equal
rows	$2x - y + z = 0$	$4x + 4y + z = 0$	columns
	$4x + y + z = 2$	$6x + 6y + z = 2.$	

- 18 Construct a 3 by 3 example that has 9 different coefficients on the left side, but rows 2 and 3 become zero in elimination. How many solutions to your system with $\mathbf{b} = (1, 10, 100)$ and how many with $\mathbf{b} = (0, 0, 0)$?
- 19 Which number q makes this system singular and which right side t gives it infinitely many solutions? Find the solution that has $z = 1$.

$$\begin{aligned}x + 4y - 2z &= 1 \\x + 7y - 6z &= 6 \\3y + qz &= t.\end{aligned}$$

- 20 Three planes can fail to have an intersection point, even if no planes are parallel. The system is singular if row 3 of A is a _____ of the first two rows. Find a third equation that can't be solved together with $x + y + z = 0$ and $x - 2y - z = 1$.
- 21 Find the pivots and the solution for both systems ($A\mathbf{x} = \mathbf{b}$ and $K\mathbf{x} = \mathbf{b}$):

$$\begin{array}{rclcl}2x + y & & = 0 & 2x - y & = 0 \\x + 2y + z & & = 0 & -x + 2y - z & = 0 \\& y + 2z + t = 0 & & -y + 2z - t = 0 \\& z + 2t = 5 & & -z + 2t = 5.\end{array}$$

- 22 If you extend Problem 21 following the 1, 2, 1 pattern or the $-1, 2, -1$ pattern, what is the fifth pivot? What is the n th pivot? K is my favorite matrix.
- 23 If elimination leads to $x + y = 1$ and $2y = 3$, find three possible original problems.
- 24 For which two numbers a will elimination fail on $A = \begin{bmatrix} a & 2 \\ a & a \end{bmatrix}$?
- 25 For which three numbers a will elimination fail to give three pivots?

$$A = \begin{bmatrix} a & 2 & 3 \\ a & a & 4 \\ a & a & a \end{bmatrix} \text{ is singular for three values of } a.$$

- 26 Look for a matrix that has row sums 4 and 8, and column sums 2 and s :

$$\text{Matrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \begin{array}{ll} a + b = 4 & a + c = 2 \\ c + d = 8 & b + d = s \end{array}$$

The four equations are solvable only if $s = \underline{\hspace{2cm}}$. Then find two different matrices that have the correct row and column sums. *Extra credit:* Write down the 4 by 4 system $A\mathbf{x} = \mathbf{b}$ with $\mathbf{x} = (a, b, c, d)$ and make A triangular by elimination.

- 27 Elimination in the usual order gives what matrix U and what solution to this “lower triangular” system? We are really solving by *forward substitution*:

$$\begin{aligned}3x &= 3 \\6x + 2y &= 8 \\9x - 2y + z &= 9.\end{aligned}$$

- 28** Create a MATLAB command $A(2, :) = \dots$ for the new row 2, to subtract 3 times row 1 from the existing row 2 if the matrix A is already known.

Challenge Problems

- 29** Find experimentally the average 1st and 2nd and 3rd pivot sizes from MATLAB's $[L, U] = \text{lu}(\text{rand}(3))$. The average size $\text{abs}(U(1, 1))$ is above $\frac{1}{2}$ because **lu** picks the largest available pivot in column 1. Here $A = \text{rand}(3)$ has random entries between 0 and 1.
- 30** If the last corner entry is $A(5, 5) = 11$ and the last pivot of A is $U(5, 5) = 4$, what different entry $A(5, 5)$ would have made A singular?
- 31** Suppose elimination takes A to U without row exchanges. Then row j of U is a combination of which rows of A ? If $Ax = \mathbf{0}$, is $Ux = \mathbf{0}$? If $Ax = b$, is $Ux = b$? If A starts out lower triangular, what is the upper triangular U ?
- 32** Start with 100 equations $Ax = \mathbf{0}$ for 100 unknowns $x = (x_1, \dots, x_{100})$. Suppose elimination reduces the 100th equation to $0 = 0$, so the system is "singular".
- (a) Elimination takes linear combinations of the rows. So this singular system has the singular property: Some linear combination of the 100 **rows** is _____.
 - (b) Singular systems $Ax = \mathbf{0}$ have infinitely many solutions. This means that some linear combination of the 100 **columns** is _____.
 - (c) Invent a 100 by 100 singular matrix with no zero entries.
 - (d) For your matrix, describe in words the row picture and the column picture of $Ax = \mathbf{0}$. Not necessary to draw 100-dimensional space.

2.3 Elimination Using Matrices

We now combine two ideas—elimination and matrices. The goal is to express all the steps of elimination (and the final result) in the clearest possible way. In a 3 by 3 example, elimination could be described in words. For larger systems, a long list of steps would be hopeless. You will see how to subtract a multiple of row j from row i —using a matrix E .

The 3 by 3 example in the previous section has the beautifully short form $Ax = b$:

$$\begin{array}{rcl} 2x_1 + 4x_2 - 2x_3 & = & 2 \\ 4x_1 + 9x_2 - 3x_3 & = & 8 \\ -2x_1 - 3x_2 + 7x_3 & = & 10 \end{array} \quad \text{is the same as} \quad \begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix}. \quad (1)$$

The nine numbers on the left go into the matrix A . That matrix not only sits beside x , it *multiplies* x . The rule for “ A times x ” is exactly chosen to yield the three equations.

Review of A times x . A matrix times a vector gives a vector. The matrix is square when the number of equations (three) matches the number of unknowns (three). Our matrix is 3 by 3. A general square matrix is n by n . Then the vector x is in n -dimensional space.

$$\text{The unknown in } \mathbb{R}^3 \text{ is } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{and the solution is } x = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}.$$

Key point: $Ax = b$ represents the row form and also the column form of the equations.

$$\text{Column form} \quad Ax = (-1) \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 9 \\ -3 \end{bmatrix} + 2 \begin{bmatrix} -2 \\ -3 \\ 7 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix} = b.$$

This rule for Ax is used so often that we express it once more for emphasis.

Ax is a combination of the columns of A . Components of x multiply those columns:

$$Ax = x_1 \text{ times (column 1)} + \cdots + x_n \text{ times (column } n).$$

When we compute the components of Ax , we use the row form of matrix multiplication. The i th component is a dot product with row i of A , which is $[a_{i1} \ a_{i2} \ \dots \ a_{in}]$. The short formula for that dot product with x uses “sigma notation”.

Components of Ax are dot products with rows of A .

$$\text{The } i\text{th component of } Ax \text{ is } a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n. \quad \text{This is } \sum_{j=1}^n a_{ij}x_j.$$

The sigma symbol $\sum$ is an instruction to add.¹ Start with $j = 1$ and stop with $j = n$. Start the sum with $a_{i1}x_1$ and stop with $a_{in}x_n$. That produces (row i) $\cdot x$.

¹Einstein shortened this even more by omitting the $\sum$. The repeated j in $a_{ij}x_j$ automatically meant addition. He also wrote the sum as $a_i^j x_j$. Not being Einstein, we include the $\sum$.

One point to repeat about matrix notation: The entry in row 1, column 1 (the top left corner) is a_{11} . The entry in row 1, column 3 is a_{13} . The entry in row 3, column 1 is a_{31} . (Row number comes before column number.) The word “entry” for a matrix corresponds to “component” for a vector. General rule: $a_{ij} = A(i, j)$ is in row i , column j .

Example 1 This matrix has $a_{ij} = 2i + j$. Then $a_{11} = 3$. Also $a_{12} = 4$ and $a_{21} = 5$. Here is Ax with numbers and letters:

$$\begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \cdot 2 + 4 \cdot 1 \\ 5 \cdot 2 + 6 \cdot 1 \end{bmatrix} \quad \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix}.$$

The first component of Ax is $6 + 4 = 10$. *A row times a column gives a dot product.*

The Matrix Form of One Elimination Step

$Ax = b$ is a convenient form for the original equation. What about the elimination steps? The first step in this example subtracts 2 times the first equation from the second equation. On the right side, 2 times the first component of b is subtracted from the second component:

First step $b = \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix}$ changes to $b_{\text{new}} = \begin{bmatrix} 2 \\ 4 \\ 10 \end{bmatrix}.$

We want to do that subtraction with a matrix! The same result $b_{\text{new}} = Eb$ is achieved when we multiply an “elimination matrix” E times b . It subtracts $2b_1$ from b_2 :

The elimination matrix is $E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

Multiplication by E subtracts 2 times row 1 from row 2. Rows 1 and 3 stay the same:

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 10 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 - 2b_1 \\ b_3 \end{bmatrix}$$

The first and third rows of E are rows from the identity matrix I . The new second component is the number 4 that appeared after the elimination step. This is $b_2 - 2b_1$.

It is easy to describe the “elementary matrices” or “elimination matrices” like this E . Start with the identity matrix I . Change one of its zeros to the multiplier $-\ell$:

The *identity matrix* has 1’s on the diagonal and otherwise 0’s. Then $Ib = b$ for all b . The *elementary matrix or elimination matrix* E_{ij} that subtracts a multiple ℓ of row j from row i has the extra nonzero entry $-\ell$ in the i, j position (still diagonal 1’s).

Example 2 The matrix E_{31} has $-\ell$ in the 3, 1 position:

$$\text{Identity } I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{Elimination } E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\ell & 0 & 1 \end{bmatrix}.$$

When you multiply I times b , you get b . But E_{31} subtracts ℓ times the first component from the third component. With $\ell = 4$ this example gives $9 - 4 = 5$:

$$Ib = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} \quad \text{and} \quad Eb = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}.$$

What about the left side of $Ax = b$? Both sides are multiplied by E_{31} . **The purpose of E_{31} is to produce a zero in the (3, 1) position of the matrix.**

The notation fits this purpose. Start with A . Apply E 's to produce zeros below the pivots (the first E is E_{21}). End with a triangular U . We now look in detail at those steps.

First a small point. The vector x stays the same. The solution is not changed by elimination. (That may be more than a small point.) It is the coefficient matrix that is changed. When we start with $Ax = b$ and multiply by E , the result is $EAx = Eb$. The new matrix EA is the result of *multiplying E times A* .

Confession The *elimination matrices* E_{ij} are great examples, but you won't see them later. They show how a matrix acts on rows. By taking several elimination steps, we will see how to *multiply matrices* (and the order of the E 's becomes important). **Products and inverses** are especially clear for E 's. It is those two ideas that the book will now use.

Matrix Multiplication

The big question is: **How do we multiply two matrices?** When the first matrix is E , we already know what to expect for EA . This particular E subtracts 2 times row 1 from row 2 of this matrix A and any matrix. The multiplier is $\ell = 2$:

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ -2 & -3 & 7 \end{bmatrix} \quad (\text{with the zero}). \quad (2)$$

This step does not change rows 1 and 3 of A . Those rows are unchanged in EA —only row 2 is different. *Twice the first row has been subtracted from the second row.* Matrix multiplication agrees with elimination—and the new system of equations is $EAx = Eb$.

EAx is simple but it involves a subtle idea. Start with $Ax = b$. Multiplying both sides by E gives $E(Ax) = Eb$. With matrix multiplication, this is also $(EA)x = Eb$. **The first was E times Ax , the second is EA times x . They are the same.** Parentheses are not needed. We just write EAx .

That rule extends to a matrix C with several column vectors like $C = [c_1 \ c_2 \ c_3]$. When multiplying EAC , you can do AC first or EA first. This is the point of an “associative law” like $3 \times (4 \times 5) = (3 \times 4) \times 5$. Multiply 3 times 20, or multiply 12 times 5. Both answers are 60. That law seems so clear that it is hard to imagine it could be false.

The “commutative law” $3 \times 4 = 4 \times 3$ looks even more obvious. But EA is usually different from AE . When E multiplies on the right, it acts on the *columns* of A .

Associative law is true

$$A(BC) = (AB)C$$

Commutative law is false

$$\text{Often } AB \neq BA$$

There is another requirement on matrix multiplication. Suppose B has only one column (this column is b). The matrix-matrix law for EB should agree with the matrix-vector law for Eb . Even more, we should be able to *multiply matrices EB a column at a time*:

If B has several columns b_1, b_2, b_3 , then the columns of EB are Eb_1, Eb_2, Eb_3 .

Matrix multiplication

$$AB = A[b_1 \ b_2 \ b_3] = [Ab_1 \ Ab_2 \ Ab_3]. \quad (3)$$

This holds true for the matrix multiplication in (2). If you multiply column 3 of A by E , you correctly get column 3 of EA :

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ -3 \\ 7 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix} \quad E(\text{column } j \text{ of } A) = \text{column } j \text{ of } EA.$$

This requirement deals with columns, while elimination is applied to rows. **The next section describes each entry of every product AB .** The beauty of matrix multiplication is that all three approaches (*rows, columns, whole matrices*) come out right.

The Matrix P_{ij} for a Row Exchange

To subtract row j from row i we use E_{ij} . To exchange or “permute” those rows we use another matrix P_{ij} (a **permutation matrix**). A row exchange is needed when zero is in the pivot position. Lower down, that pivot column may contain a nonzero. By exchanging the two rows, we have a pivot and elimination goes forward.

What matrix P_{23} exchanges row 2 with row 3? We can find it by exchanging rows of the identity matrix I :

$$\text{Permutation matrix} \quad P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

This is a **row exchange matrix**. Multiplying by P_{23} exchanges components 2 and 3 of any column vector. Therefore it also exchanges rows 2 and 3 of any matrix:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ 0 & 0 & 3 \\ 0 & 6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 3 \end{bmatrix}.$$

On the right, P_{23} is doing what it was created for. With zero in the second pivot position and “6” below it, the exchange puts 6 into the pivot.

Matrices *act*. They don't just sit there. We will soon meet other permutation matrices, which can change the order of several rows. Rows 1, 2, 3 can be moved to 3, 1, 2. Our P_{23} is one particular permutation matrix—it exchanges rows 2 and 3.

Row Exchange Matrix P_{ij} is the identity matrix with rows i and j reversed. When this “permutation matrix” P_{ij} multiplies a matrix, it exchanges rows i and j .

To exchange equations 1 and 3 multiply by $P_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.

Usually row exchanges are not required. The odds are good that elimination uses only the E_{ij} . But the P_{ij} are ready if needed, to move a pivot up to the diagonal.

The Augmented Matrix

This book eventually goes far beyond elimination. Matrices have all kinds of practical applications, in which they are multiplied. Our best starting point was a square E times a square A , because we met this in elimination—and we know what answer to expect for EA . The next step is to allow a *rectangular matrix*. It still comes from our original equations, but now it includes the right side b .

Key idea: Elimination does the same row operations to A and to b . *We can include b as an extra column and follow it through elimination.* The matrix A is enlarged or “augmented” by the extra column b :

$$\text{Augmented matrix} \quad [A \ b] = \begin{bmatrix} 2 & 4 & -2 & 2 \\ 4 & 9 & -3 & 8 \\ -2 & -3 & 7 & 10 \end{bmatrix}.$$

Elimination acts on whole rows of this matrix. The left side and right side are both multiplied by E , to subtract 2 times equation 1 from equation 2. With $[A \ b]$ those steps happen together:

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & -2 & 2 \\ 4 & 9 & -3 & 8 \\ -2 & -3 & 7 & 10 \end{bmatrix} = \begin{bmatrix} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 4 \\ -2 & -3 & 7 & 10 \end{bmatrix}.$$

The new second row contains 0, 1, 1, 4. The new second equation is $x_2 + x_3 = 4$. Matrix multiplication works by rows and at the same time by columns:

ROWS Each row of E acts on $[A \ b]$ to give a row of $[EA \ Eb]$.

COLUMNS E acts on each column of $[A \ b]$ to give a column of $[EA \ Eb]$.

Notice again that word “acts.” This is essential. Matrices do something! The matrix A acts on x to produce b . The matrix E operates on A to give EA . The whole process of elimination is a sequence of row operations, alias matrix multiplications. A goes to $E_{21}A$ which goes to $E_{31}E_{21}A$. Finally $E_{32}E_{31}E_{21}A$ is a triangular matrix.

The right side is included in the augmented matrix. The end result is a triangular system of equations. We stop for exercises on multiplication by E , before writing down the rules for all matrix multiplications (including block multiplication).

■ REVIEW OF THE KEY IDEAS ■

1. $Ax = x_1$ times column 1 $+\cdots+x_n$ times column n . And $(Ax)_i = \sum_{j=1}^n a_{ij}x_j$.
2. Identity matrix $= I$, elimination matrix $= E_{ij}$ using ℓ_{ij} , exchange matrix $= P_{ij}$.
3. Multiplying $Ax = b$ by E_{21} subtracts a multiple ℓ_{21} of equation 1 from equation 2. The number $-\ell_{21}$ is the $(2, 1)$ entry of the elimination matrix E_{21} .
4. For the augmented matrix $[A \ b]$, that elimination step gives $[E_{21}A \ E_{21}b]$.
5. When A multiplies any matrix B , it multiplies each column of B separately.

■ WORKED EXAMPLES ■

2.3 A What 3 by 3 matrix E_{21} subtracts 4 times row 1 from row 2? What matrix P_{32} exchanges row 2 and row 3? If you multiply A on the *right* instead of the left, describe the results AE_{21} and AP_{32} .

Solution By doing those operations on the identity matrix I , we find

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad P_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Multiplying by E_{21} on the right side will subtract 4 times **column 2** from **column 1**. Multiplying by P_{32} on the right will exchange **columns 2 and 3**.

2.3 B Write down the augmented matrix $[A \ b]$ with an extra column:

$$\begin{aligned} x + 2y + 2z &= 1 \\ 4x + 8y + 9z &= 3 \\ 3y + 2z &= 1 \end{aligned}$$

Apply E_{21} and then P_{32} to reach a triangular system. Solve by back substitution. What combined matrix $P_{32}E_{21}$ will do both steps at once?

Solution E_{21} removes the 4 in column 1. But zero appears in column 2:

$$[A \ b] = \begin{bmatrix} 1 & 2 & 2 & 1 \\ 4 & 8 & 9 & 3 \\ 0 & 3 & 2 & 1 \end{bmatrix} \quad \text{and} \quad E_{21}[A \ b] = \begin{bmatrix} 1 & 2 & 2 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 3 & 2 & 1 \end{bmatrix}$$

Now P_{32} exchanges rows 2 and 3. Back substitution produces z then y and x .

$$P_{32} E_{21}[A \ b] = \begin{bmatrix} 1 & 2 & 2 & 1 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

For the matrix $P_{32} E_{21}$ that does both steps at once, *apply* P_{32} to E_{21} .

One matrix
Both steps $P_{32} E_{21} = \text{exchange the rows of } E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ -4 & 1 & 0 \end{bmatrix}.$

2.3 C Multiply these matrices in two ways. First, rows of A times columns of B . Second, *columns of A times rows of B* . That unusual way produces two matrices that add to AB . How many separate ordinary multiplications are needed?

Both ways $AB = \begin{bmatrix} 3 & 4 \\ 1 & 5 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 16 \\ 7 & 9 \\ 4 & 8 \end{bmatrix}$

Solution Rows of A times columns of B are dot products of vectors:

$$(\text{row } 1) \cdot (\text{column } 1) = \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 10 \quad \text{is the } (1, 1) \text{ entry of } AB$$

$$(\text{row } 2) \cdot (\text{column } 1) = \begin{bmatrix} 1 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 7 \quad \text{is the } (2, 1) \text{ entry of } AB$$

We need 6 dot products, 2 multiplications each, 12 in all ($3 \cdot 2 \cdot 2$). The same AB comes from *columns of A times rows of B* . A column times a row is a matrix.

$$AB = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 4 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 2 & 4 \\ 4 & 8 \end{bmatrix} + \begin{bmatrix} 4 & 4 \\ 5 & 5 \\ 0 & 0 \end{bmatrix}$$

Problem Set 2.3

Problems 1–15 are about elimination matrices.

- 1 Write down the 3 by 3 matrices that produce these elimination steps:
 - (a) E_{21} subtracts 5 times row 1 from row 2.
 - (b) E_{32} subtracts -7 times row 2 from row 3.
 - (c) P exchanges rows 1 and 2, then rows 2 and 3.
- 2 In Problem 1, applying E_{21} and then E_{32} to $\mathbf{b} = (1, 0, 0)$ gives $E_{32}E_{21}\mathbf{b} = \underline{\hspace{2cm}}$. Applying E_{32} before E_{21} gives $E_{21}E_{32}\mathbf{b} = \underline{\hspace{2cm}}$. When E_{32} comes first, row $\underline{\hspace{2cm}}$ feels no effect from row $\underline{\hspace{2cm}}$.
- 3 Which three matrices E_{21} , E_{31} , E_{32} put A into triangular form U ?

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix} \quad \text{and} \quad E_{32}E_{31}E_{21}A = U.$$

Multiply those E 's to get one matrix M that does elimination: $MA = U$.

- 4 Include $\mathbf{b} = (1, 0, 0)$ as a fourth column in Problem 3 to produce $[A \ \mathbf{b}]$. Carry out the elimination steps on this augmented matrix to solve $A\mathbf{x} = \mathbf{b}$.
- 5 Suppose $a_{33} = 7$ and the third pivot is 5. If you change a_{33} to 11, the third pivot is $\underline{\hspace{2cm}}$. If you change a_{33} to $\underline{\hspace{2cm}}$, there is no third pivot.
- 6 If every column of A is a multiple of $(1, 1, 1)$, then $A\mathbf{x}$ is always a multiple of $(1, 1, 1)$. Do a 3 by 3 example. How many pivots are produced by elimination?
- 7 Suppose E subtracts 7 times row 1 from row 3.
 - (a) To *invert* that step you should $\underline{\hspace{2cm}}$ 7 times row $\underline{\hspace{2cm}}$ to row $\underline{\hspace{2cm}}$.
 - (b) What “inverse matrix” E^{-1} takes that reverse step (so $E^{-1}E = I$)?
 - (c) If the reverse step is applied first (and then E) show that $EE^{-1} = I$.
- 8 The *determinant* of $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\det M = ad - bc$. Subtract ℓ times row 1 from row 2 to produce a new M^* . Show that $\det M^* = \det M$ for every ℓ . When $\ell = c/a$, the product of pivots equals the determinant: $(a)(d - \ell b)$ equals $ad - bc$.
- 9
 - (a) E_{21} subtracts row 1 from row 2 and then P_{23} exchanges rows 2 and 3. What matrix $M = P_{23}E_{21}$ does both steps at once?
 - (b) P_{23} exchanges rows 2 and 3 and then E_{31} subtracts row 1 from row 3. What matrix $M = E_{31}P_{23}$ does both steps at once? Explain why the M 's are the same but the E 's are different.

- 10 (a) What 3 by 3 matrix E_{13} will add row 3 to row 1?
 (b) What matrix adds row 1 to row 3 and *at the same time* row 3 to row 1?
 (c) What matrix adds row 1 to row 3 and *then* adds row 3 to row 1?
- 11 Create a matrix that has $a_{11} = a_{22} = a_{33} = 1$ but elimination produces two negative pivots without row exchanges. (The first pivot is 1.)
- 12 Multiply these matrices:

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 1 \\ 1 & 4 & 0 \end{bmatrix}.$$

- 13 Explain these facts. If the third column of B is all zero, the third column of EB is all zero (for any E). If the third *row* of B is all zero, the third row of EB might *not* be zero.
- 14 This 4 by 4 matrix will need elimination matrices E_{21} and E_{32} and E_{43} . What are those matrices?

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}.$$

- 15 Write down the 3 by 3 matrix that has $a_{ij} = 2i - 3j$. This matrix has $a_{32} = 0$, but elimination still needs E_{32} to produce a zero in the 3, 2 position. Which previous step destroys the original zero and what is E_{32} ?

Problems 16–23 are about creating and multiplying matrices.

- 16 Write these ancient problems in a 2 by 2 matrix form $Ax = b$ and solve them:
- (a) X is twice as old as Y and their ages add to 33.
 (b) $(x, y) = (2, 5)$ and $(3, 7)$ lie on the line $y = mx + c$. Find m and c .
- 17 The parabola $y = a + bx + cx^2$ goes through the points $(x, y) = (1, 4)$ and $(2, 8)$ and $(3, 14)$. Find and solve a matrix equation for the unknowns (a, b, c) .
- 18 Multiply these matrices in the orders EF and FE :

$$E = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & 0 & 1 \end{bmatrix} \quad F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c & 1 \end{bmatrix}.$$

Also compute $E^2 = EE$ and $F^3 = FFF$. You can guess F^{100} .

- 19 Multiply these row exchange matrices in the orders PQ and QP and P^2 :

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad Q = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Find another non-diagonal matrix whose square is $M^2 = I$.

- 20 (a) Suppose all columns of B are the same. Then all columns of EB are the same, because each one is E times _____.
 (b) Suppose all rows of B are $[1 \ 2 \ 4]$. Show by example that all rows of EB are *not* $[1 \ 2 \ 4]$. It is true that those rows are _____.
- 21 If E adds row 1 to row 2 and F adds row 2 to row 1, does EF equal FE ?
- 22 The entries of A and x are a_{ij} and x_j . So the first component of Ax is $\sum a_{1j}x_j = a_{11}x_1 + \cdots + a_{1n}x_n$. If E_{21} subtracts row 1 from row 2, write a formula for
 (a) the third component of Ax
 (b) the $(2, 1)$ entry of $E_{21}A$
 (c) the $(2, 1)$ entry of $E_{21}(E_{21}A)$
 (d) the first component of $E_{21}Ax$.
- 23 The elimination matrix $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ subtracts 2 times row 1 of A from row 2 of A . The result is EA . What is the effect of $E(EA)$? In the opposite order AE , we are subtracting 2 times _____ of A from _____. (Do examples.)

Problems 24–27 include the column b in the augmented matrix $[A \ b]$.

- 24 Apply elimination to the 2 by 3 augmented matrix $[A \ b]$. What is the triangular system $Ux = c$? What is the solution x ?

$$Ax = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 17 \end{bmatrix}.$$

- 25 Apply elimination to the 3 by 4 augmented matrix $[A \ b]$. How do you know this system has no solution? Change the last number 6 so there *is* a solution.

$$Ax = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix}.$$

- 26 The equations $Ax = b$ and $Ax^* = b^*$ have the same matrix A . What double augmented matrix should you use in elimination to solve both equations at once?

Solve both of these equations by working on a 2 by 4 matrix:

$$\begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

- 27 Choose the numbers a, b, c, d in this augmented matrix so that there is (a) no solution (b) infinitely many solutions.

$$[A \ b] = \begin{bmatrix} 1 & 2 & 3 & a \\ 0 & 4 & 5 & b \\ 0 & 0 & d & c \end{bmatrix}$$

Which of the numbers a, b, c , or d have no effect on the solvability?

- 28 If $AB = I$ and $BC = I$ use the associative law to prove $A = C$.

Challenge Problems

- 29 Find the triangular matrix E that reduces “Pascal’s matrix” to a smaller Pascal:

Eliminate column 1

$$E \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix}.$$

Which matrix M (multiplying several E ’s) reduces Pascal all the way to I ? Pascal’s triangular matrix is exceptional, all of its multipliers are $\ell_{ij} = 1$.

- 30 Write $M = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}$ as a product of many factors $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.
- (a) What matrix E subtracts row 1 from row 2 to make row 2 of EM smaller?
 - (b) What matrix F subtracts row 2 of EM from row 1 to reduce row 1 of FEM ?
 - (c) Continue E ’s and F ’s until (many E ’s and F ’s) times (M) is (A or B).
 - (d) E and F are the inverses of A and B ! Moving all E ’s and F ’s to the right side will give you the desired result $M = \text{product of } A\text{'s and } B\text{'s}$.

This is possible for integer matrices $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} > 0$ that have $ad - bc = 1$.

- 31 Find elimination matrices E_{21} then E_{32} then E_{43} to change K into U :

$$E_{43} E_{32} E_{21} \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & 0 & 4/3 & -1 \\ 0 & 0 & 0 & 5/4 \end{bmatrix}.$$

Apply those three steps to the identity matrix I , to multiply $E_{43}E_{32}E_{21}$.

2.4 Rules for Matrix Operations

I will start with basic facts. A matrix is a rectangular array of numbers or “entries”. When A has m rows and n columns, it is an “ m by n ” matrix. Matrices can be added if their shapes are the same. They can be multiplied by any constant c . Here are examples of $A + B$ and $2A$, for 3 by 2 matrices:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 2 \\ 4 & 4 \\ 9 & 9 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 7 & 8 \\ 9 & 9 \end{bmatrix} \quad \text{and} \quad 2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \\ 0 & 0 \end{bmatrix}.$$

Matrices are added exactly as vectors are—one entry at a time. We could even regard a column vector as a matrix with only one column (so $n = 1$). The matrix $-A$ comes from multiplication by $c = -1$ (reversing all the signs). Adding A to $-A$ leaves the *zero matrix*, with all entries zero. All this is only common sense.

The entry in row i and column j is called a_{ij} or $A(i, j)$. The n entries along the first row are $a_{11}, a_{12}, \dots, a_{1n}$. The lower left entry in the matrix is a_{m1} and the lower right is a_{mn} . The row number i goes from 1 to m . The column number j goes from 1 to n .

Matrix addition is easy. The serious question is **matrix multiplication**. When can we multiply A times B , and what is the product AB ? We cannot multiply when A and B are 3 by 2. They don’t pass the following test:

To multiply AB : *If A has n columns, B must have n rows.*

When A is 3 by 2, the matrix B can be 2 by 1 (a vector) or 2 by 2 (square) or 2 by 20. **Every column of B is multiplied by A .** I will begin matrix multiplication the *dot product* way, and then return to this *column* way: A times columns of B . The most important rule is that **AB times C equals A times BC** . A Challenge Problem will prove this.

Suppose A is m by n and B is n by p . We can multiply. The product AB is m by p .

$$(m \times n)(n \times p) = (m \times p) \quad \begin{bmatrix} m \text{ rows} \\ n \text{ columns} \end{bmatrix} \begin{bmatrix} n \text{ rows} \\ p \text{ columns} \end{bmatrix} = \begin{bmatrix} m \text{ rows} \\ p \text{ columns} \end{bmatrix}.$$

A row times a column is an extreme case. Then 1 by n multiplies n by 1. The result is 1 by 1. That single number is the “dot product”.

In every case AB is filled with dot products. For the top corner, the $(1, 1)$ entry of AB is (row 1 of A) $\cdot$ (column 1 of B). To multiply matrices, take the dot product of **each row of A with each column of B** .

The entry in row i and column j of AB is (row i of A) $\cdot$ (column j of B).

Figure 2.8 picks out the second row ($i = 2$) of a 4 by 5 matrix A . It picks out the third column ($j = 3$) of a 5 by 6 matrix B . Their dot product goes into row 2 and column 3 of AB . The matrix AB has *as many rows as A* (4 rows), and *as many columns as B* .

$$\begin{array}{ccc}
 \begin{bmatrix} * & & & & \\ a_{i1} & a_{i2} & \cdots & a_{i5} & \\ * & & & & \\ * & & & & \end{bmatrix} & \begin{bmatrix} * & * & b_{1j} & * & * & * \\ & & b_{2j} & & & \\ & & \vdots & & & \\ & & b_{5j} & & & \end{bmatrix} & = & \begin{bmatrix} & & * & & & \\ * & * & (AB)_{ij} & * & * & * \\ & & * & & & \\ & & * & & & \end{bmatrix} \\
 A \text{ is } 4 \text{ by } 5 & B \text{ is } 5 \text{ by } 6 & & AB \text{ is } 4 \text{ by } 6
 \end{array}$$

Figure 2.8: Here $i = 2$ and $j = 3$. Then $(AB)_{23}$ is (row 2) $\cdot$ (column 3) $= \sum a_{2k}b_{k3}$.

Example 1 Square matrices can be multiplied if and only if they have the same size:

$$\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 0 \end{bmatrix}.$$

The first dot product is $1 \cdot 2 + 1 \cdot 3 = 5$. Three more dot products give 6, 1, and 0. Each dot product requires two multiplications—thus eight in all.

If A and B are n by n , so is AB . It contains n^2 dot products, row of A times column of B . Each dot product needs n multiplications, so *the computation of AB uses n^3 separate multiplications*. For $n = 100$ we multiply a million times. For $n = 2$ we have $n^3 = 8$.

Mathematicians thought until recently that AB absolutely needed $2^3 = 8$ multiplications. Then somebody found a way to do it with 7 (and extra additions). By breaking n by n matrices into 2 by 2 blocks, this idea also reduced the count for large matrices. Instead of n^3 it went below $n^{2.8}$, and the exponent keeps falling.¹ The best at this moment is $n^{2.376}$. But the algorithm is so awkward that scientific computing is done the regular way: n^2 dot products in AB , and n multiplications for each one.

Example 2 Suppose A is a row vector (1 by 3) and B is a column vector (3 by 1). Then AB is 1 by 1 (only one entry, the dot product). On the other hand B times A (*a column times a row*) is a full 3 by 3 matrix. This multiplication is allowed!

$$\begin{array}{l}
 \text{Column times row} \\
 (n \times 1)(1 \times n) = (n \times n)
 \end{array}
 \quad
 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} [1 \ 2 \ 3] = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

A row times a column is an “inner” product—that is another name for dot product. A column times a row is an “outer” product. These are extreme cases of matrix multiplication.

Rows and Columns of AB

In the big picture, A multiplies each column of B . The result is a column of AB . In that column, we are combining the columns of A . *Each column of AB is a combination of*

¹Maybe 2.376 will drop to 2. No other number looks special, but no change for 10 years.

the columns of A. That is the column picture of matrix multiplication:

$$\text{Matrix } A \text{ times column of } B \quad A[b_1 \cdots b_p] = [Ab_1 \cdots Ab_p].$$

The row picture is reversed. Each row of A multiplies the whole matrix B . The result is a row of AB . It is a combination of the rows of B :

$$\text{Row times matrix} \quad [\text{row } i \text{ of } A] \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = [\text{row } i \text{ of } AB].$$

We see row operations in elimination (E times A). We see columns in A times x . The “row-column picture” has the dot products of rows with columns. Believe it or not, *there is also a column-row picture*. Not everybody knows that columns $1, \dots, n$ of A multiply rows $1, \dots, n$ of B and add up to the same answer AB . Worked Example 2.3 C had numbers for $n = 2$. *Example 3 will show how to multiply AB using columns times rows*.

The Laws for Matrix Operations

May I put on record six laws that matrices do obey, while emphasizing an equation they don't obey? The matrices can be square or rectangular, and the laws involving $A + B$ are all simple and all obeyed. Here are three addition laws:

$$\begin{aligned} A + B &= B + A && \text{(commutative law)} \\ c(A + B) &= cA + cB && \text{(distributive law)} \\ A + (B + C) &= (A + B) + C && \text{(associative law).} \end{aligned}$$

Three more laws hold for multiplication, but $AB = BA$ is not one of them:

$$\begin{aligned} AB &\neq BA && \text{(the commutative “law” is usually broken)} \\ C(A + B) &= CA + CB && \text{(distributive law from the left)} \\ (A + B)C &= AC + BC && \text{(distributive law from the right)} \\ A(BC) &= (AB)C && \text{(associative law for } ABC) \text{ (parentheses not needed).} \end{aligned}$$

When A and B are not square, AB is a different size from BA . These matrices can't be equal—even if both multiplications are allowed. For square matrices, almost any example shows that AB is different from BA :

$$AB = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{but} \quad BA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

It is true that $AI = IA$. All square matrices commute with I and also with cI . Only these matrices cI commute with all other matrices.

The law $A(B + C) = AB + AC$ is proved a column at a time. Start with $A(b + c) = Ab + Ac$ for the first column. That is the key to everything—*linearity*. Say no more.

The law $A(BC) = (AB)C$ means that you can multiply BC first or else AB first. The direct proof is sort of awkward (Problem 37) but this law is extremely useful. We highlighted it above; it is the key to the way we multiply matrices.